

05/10/2026 - ATIVIDADE

1.

$$a) \begin{cases} 2x + y = 7 \\ x - 3y = -5 \end{cases}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \end{bmatrix}$$

$$b) \begin{cases} x + 2y - z = 2 \\ 3x - y + 2z = 9 \\ 2x + y + z = 7 \end{cases}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \\ 7 \end{bmatrix}$$

2.

$$\begin{cases} 2x + y - z = 8 \\ -3x - y + 2z = -11 \\ -2x + y + 2z = -3 \end{cases}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right] \quad L_1 \cdot 1 + L_3 =$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ 0 & 2 & 1 & 5 \end{array} \right] \quad L_1 \cdot \frac{3}{2} + L_2 =$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{11}{2} \\ 0 & 2 & 1 & 5 \end{array} \right] \quad L_2 \cdot 2 =$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 5 \end{array} \right] \quad L_2 \cdot (-2) + L_3 =$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

$$\begin{cases} 2x + y - z = 8 \Rightarrow 2x = 4 \Rightarrow \boxed{x = 2} \\ y + z = 2 \Rightarrow \boxed{y = 3} \\ -z = 1 \Rightarrow \boxed{z = -1} \end{cases}$$

3.

$$a) \begin{cases} x + y = 3 \\ 2x + 2y = 6 \end{cases}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 2 & 6 \end{array} \right] \xrightarrow{L_1 \cdot (-2) + L_2} =$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & 0 \end{array} \right]$$

Resta $x + y = 3 \Rightarrow x = 3 - y$, tomando

y como um a qualquer; $a \in \mathbb{R}$, temos infinitas soluções.

$$b) \begin{cases} x + y = 3 \\ 2x + 2y = 7 \end{cases}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 2 & 7 \end{array} \right] \quad L_1 \cdot (-2) + L_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & 1 \end{array} \right]$$

como a última linha diz $0=1$, não há solução.

$$c) \begin{cases} x + y = 3 \\ 2x - y = 0 \end{cases}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right] \quad L_1 \cdot (-2) + L_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array} \right] \Rightarrow \begin{cases} x + y = 3 \Rightarrow x = 1 \\ -3y = -6 \Rightarrow y = 2 \end{cases}$$

Então, há apenas uma solução, $x=1$ e $y=2$.

5.

$$a) \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 10 \\ 10 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 4 & -1 & 0 & 15 \\ -1 & 4 & -1 & 10 \\ 0 & -1 & 3 & 10 \end{array} \right] \quad \frac{L_1}{4} + L_2 =$$

$$\left[\begin{array}{ccc|c} 4 & -1 & 0 & 15 \\ 0 & 15/4 & -1 & 55/4 \\ 0 & -1 & 3 & 10 \end{array} \right] \quad L_3 \cdot \frac{15}{4} + L_2 =$$

$$\left[\begin{array}{ccc|c} 4 & -1 & 0 & 15 \\ 0 & 0 & 41/4 & 205/4 \\ 0 & -1 & 3 & 10 \end{array} \right] \quad L_2 \cdot 4 = \left[\begin{array}{ccc|c} 4 & -1 & 0 & 15 \\ 0 & 0 & 41 & 205 \\ 0 & -1 & 3 & 10 \end{array} \right]$$

$$\left\{ \begin{array}{l} 4T_1 - T_2 = 15 \Rightarrow 4T_1 = 20 \Rightarrow T_1 = 5 \\ T_3 = 205/4 = 5 \\ -T_2 + 3T_3 = 10 \Rightarrow -T_2 = -5 \Rightarrow T_2 = 5 \end{array} \right.$$